

Gradient-Sign Masking for Task Vector Transport Across Pre-Trained Models

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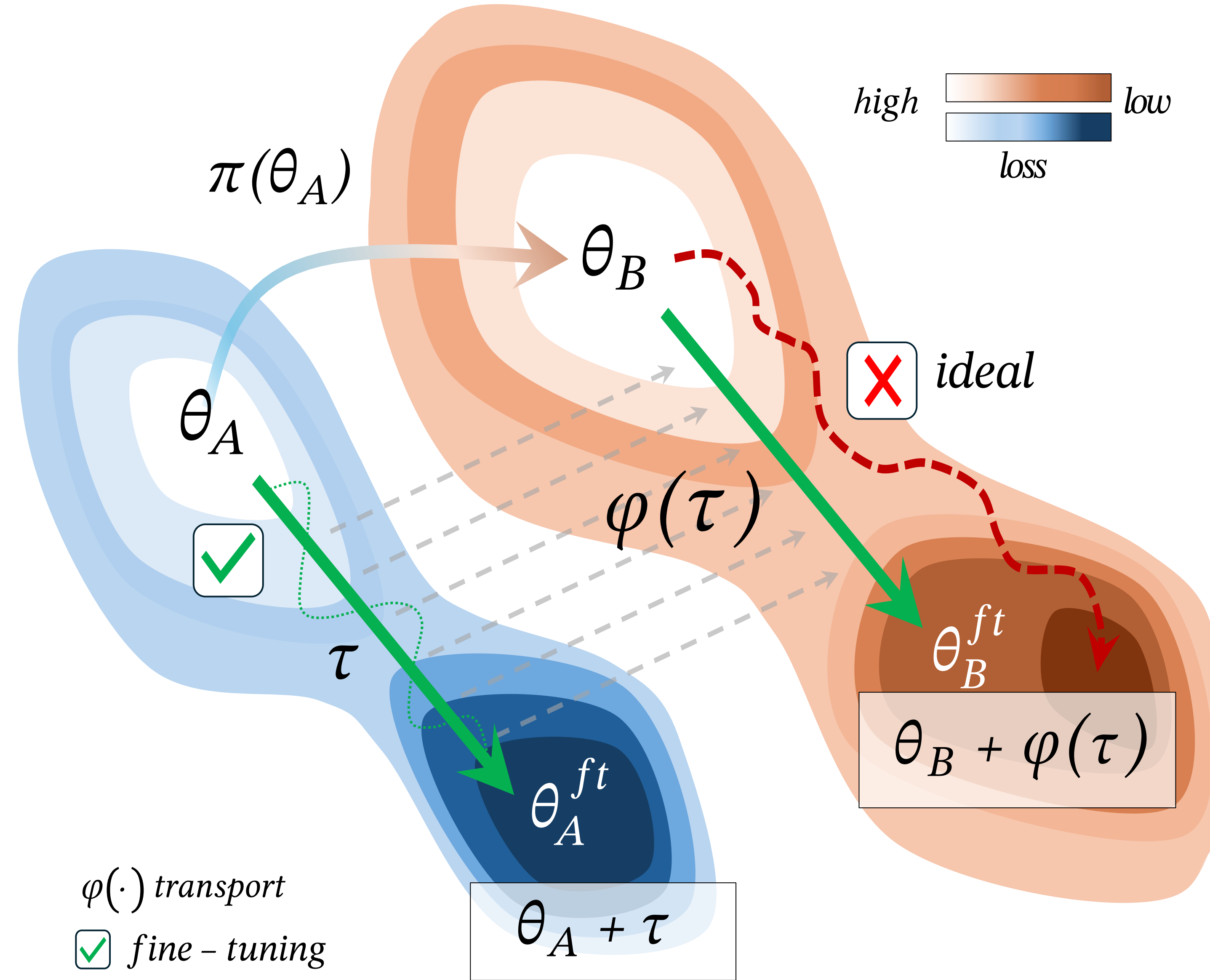
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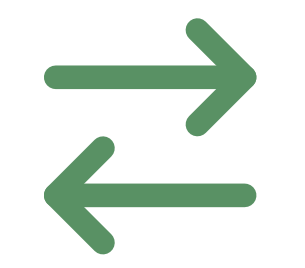
Problem

Upgrading to a new foundation model typically requires costly re-finetuning from scratch.

Rebasin: Transporting the previous model's task vector ($\tau_A = \theta_A^{ft} - \theta_A$) recycles past adaptation without training from scratch.



Background



The Transfer Failure: Naive transfer across different pre-trained models fails ($\theta_B + \tau_A$). Misaligned parameter spaces **introduce harmful directions** that increase the target model loss.



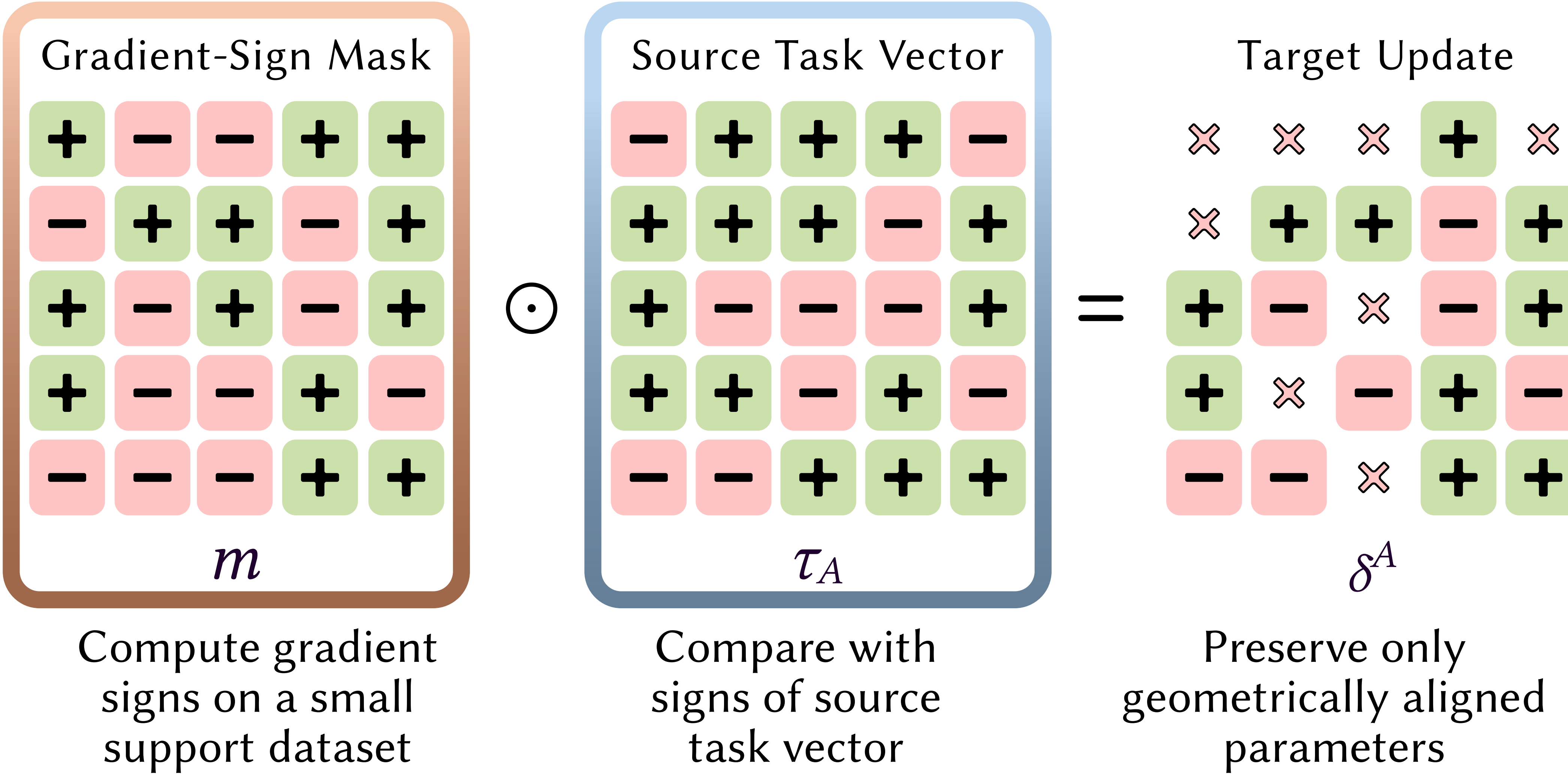
Previous Work: *TransFusion*^[1] **aligns models** via explicit weight permutations to **enable task vector transport** across distinct Transformer backbones.



The Limitation: Transported vectors **are not guaranteed** to align with the new model's local loss geometry, allowing misaligned directions to **degrade performance**.

TL;DR: Effective transfer for **rebasin** requires **alignment** with the target model's local loss geometry. **Gradfix** uses the **sign** of the target's zero-shot loss **gradient** as a **reliable proxy** for the true descent direction.

Method



Descent Guarantee

First-order approximation shows how an update affects the target model's loss.

$$\mathcal{L}(\theta_B + \delta^A) \approx \mathcal{L}(\theta_B) + g^\top \delta^A, \quad \text{where } g = \nabla_{\theta} \mathcal{L}(\theta_B). \quad [1]$$

Masking keeps components aligned with anti-gradient.

$$g_i \cdot (m_i \tau_{A,i}) = \begin{cases} -|g_i| |\tau_{A,i}|, & \text{if } \text{sign}(\tau_{A,i}) = \text{sign}(-g_i), \\ 0, & \text{otherwise,} \end{cases} \quad [2]$$

$$g^\top \delta^A = -\alpha \sum_i m_i |g_i| |\tau_{A,i}| \leq 0. \quad [3]$$

Algorithm

Require: Source model θ_A , θ_A^{ft} , target model θ_B , target data subset $\mathcal{D}_s$, scaling α .

Ensure: Transported model θ_B^{trans}

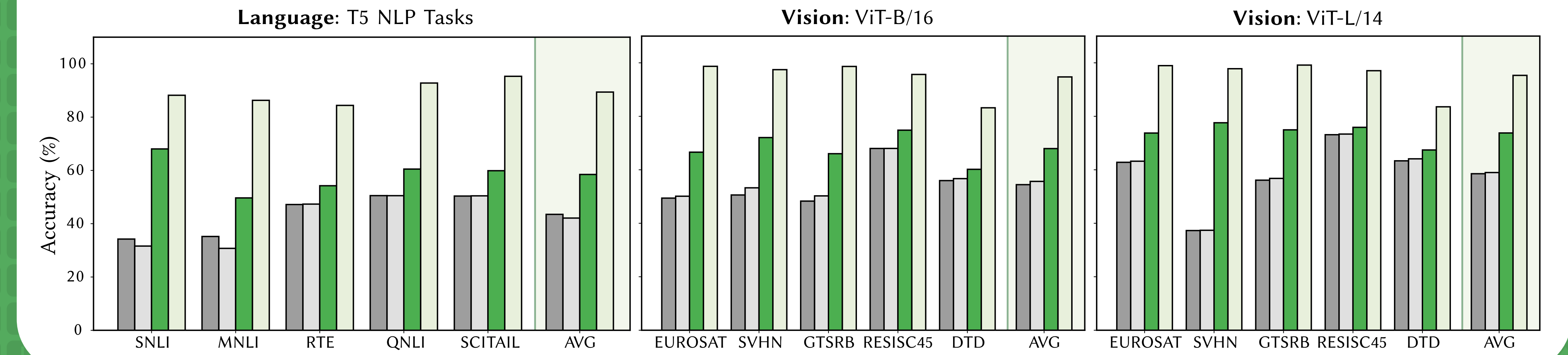
- 1: Compute source task vector: $\tau_A \leftarrow \theta_A^{ft} - \theta_A$
- 2: **for** $(x_n, y_n) \in \mathcal{D}_s$ **do**
- 3: $g^{(n)} \leftarrow \nabla_{\theta} \ell(f_{\theta_B}(x_n), y_n)$
- 4: Compute anti-gradient signs $\hat{s}_i$ by majority voting
- 5: Build gradient-sign mask: $m_i \leftarrow \{\text{sign}(\tau_{A,i}) = \hat{s}_i\}$
- 6: Compute transported update: $\delta^A \leftarrow \alpha (m \odot \tau_A)$
- 7: **return** $\theta_B^{trans} \leftarrow \theta_B + \delta^A$

Experiments

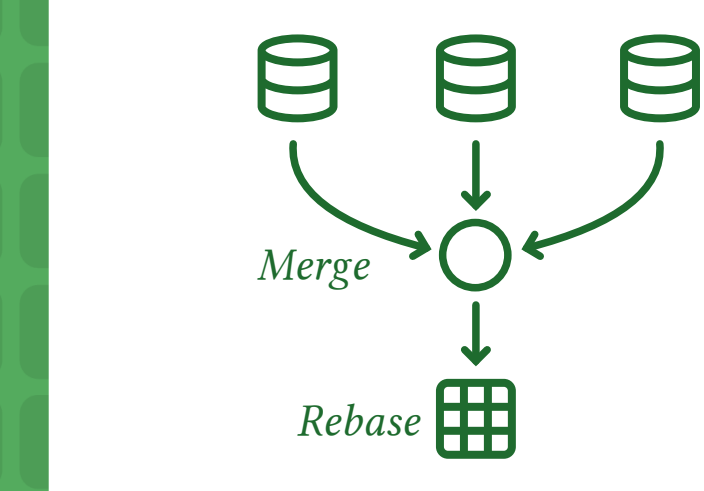
Cross-dataset Transport Performance

Enhanced transportation of task vectors from A to B (+28.41% on average!)

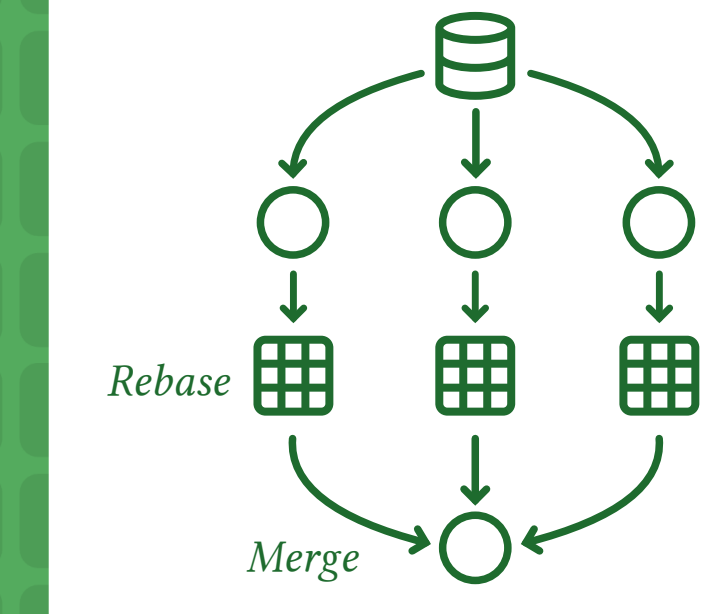
GradFix Transfusion
 θ_B zero-shot θ_B fine-tune



Model Merging & Rebasin



Multi-Task: Merge first! Preserves task synergies before filtering.

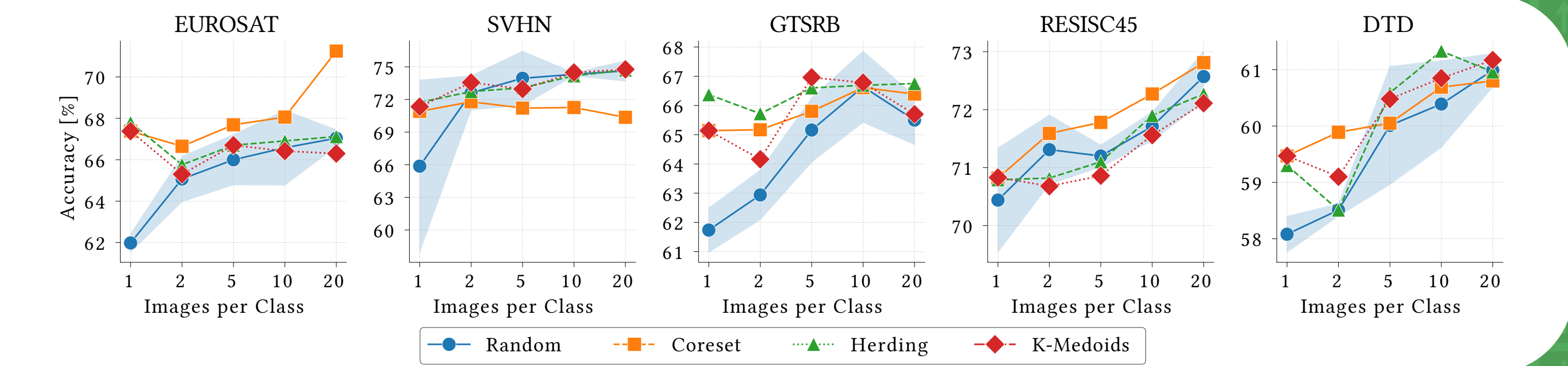


Multi-Source: Mask first! Prevents cross-model collapse during aggregation.

Pipeline	EUROSAT	SVHN	GTSRB	RESISC45	DTD	AVG
θ_B zero-shot	49.41	50.58	48.29	67.98	55.96	54.44
Multi-task ($\{\tau_{A,j}\}_{j=1}^K \rightarrow \theta_B$)						
$\theta_B + \tau_{A,j}$	49.58	50.84	49.31	67.87	56.27	54.77
$\theta_B + \delta_j^A$	65.07	70.19	64.33	71.42	58.51	65.90
Task Arithmetic						
Baseline	49.31	50.99	48.73	68.05	56.54	54.73
Mask-then-Merge	55.90	71.56	59.65	71.40	57.66	63.23
Merge-then-Mask	65.37	72.10	59.55	71.16	57.07	65.05
TIES-Merging						
Baseline	49.15	50.75	48.95	67.97	56.54	54.67
Mask-then-Merge	50.41	64.28	54.05	69.51	57.13	59.08
Merge-then-Mask	65.62	72.42	62.73	71.57	57.77	66.02
Multi-source ($\{\tau_{A,k}\}_{k=1}^K \rightarrow \theta_B$)						
Task Arithmetic						
Baseline	36.94	35.09	30.77	50.54	44.73	39.61
Merge-then-Mask	12.52	15.94	4.20	3.97	2.93	7.91
Mask-then-Merge	65.96	72.97	65.80	71.30	61.01	67.41
TIES-Merging						
Baseline	12.69	10.39	2.85	5.81	16.33	9.61
Merge-then-Mask	14.46	15.94	3.09	3.35	2.66	7.90
Mask-then-Merge	65.17	72.58	65.36	71.40	61.12	67.13

Selection Strategies

Keep It Simple: Random subset selection rivals expensive, feature-based sampling.



Gradient Signs Aggregation

Majority Voting wins! It is much more resilient to α changes than the mean.

